Chapter 1

1) functions

Sets
- real numbers (ℝ)
- integers (ℤ)
- rational (ℚ)

finite ex: 5/8 = 0.375
repeating ex: 1/3 = 0.142857142857...
infinite ex: π = 3.14159265...

decimal: |a| = distance from the origin = $\begin{cases} a & \text{if } a \geq 0 \\ -a & \text{if } a < 0 \end{cases}$

$|a| = |-a|$

$|ab| = |a||b|$

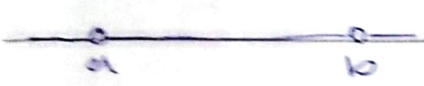
distance between a and b is $|b-a|$

$|a+b| \neq |a| + |b|$

$|a+b| < |a| + |b|$



closed interval $[a,b] = \{x \in \mathbb{R} : a \leq x \leq b\}$



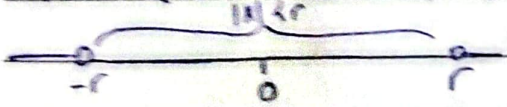
open interval $(a,b) = \{x : a < x < b\}$



half-open → $[a,b) = \{x : a \leq x < b\}$
$(a,b] = \{x : a < x \leq b\}$

$(-r,r)$ is described by the inequality $|x| < r$

$|x| < r = -r < x < r = x \in (-r,r)$



[r = radius]
[c = center]

$c = \frac{1}{2}(a+b)$ $\quad r = \frac{1}{2}(a-b)$

Describe the set: $S = \{x : |\frac{1}{2}x - 3| \geq 4\}$ {E.x}

Solution:-

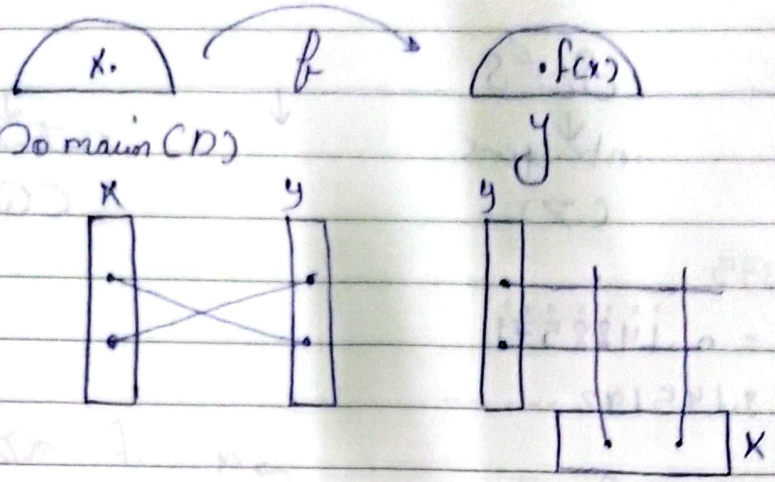
$|\frac{1}{2}x - 3| \leq 4$

$-4 \leq \frac{1}{2}x - 3 \leq 4$

$-1 \leq \frac{1}{2}x \leq 7$

$-2 \leq x \leq 14$

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$y = f(x)$

D = domain set (x)

y = set containing the range

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